MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER PRACTICAL EXAMINATION

MG3DSEPHY207 – Properties of Matter

Duration:02hrs Maximum Marks: 35

Record:5 marks

Attempt the question marked 'X'. Change of question is not allowed: 30 marks

1. Determine the poisson's ratio of rubber. [A,An,E][7]
2. Determine the rigidity modulus of a metalic rod using Static torsion method. [A,An,E][7]
3. Determine the rigidity modulus of a given wire using Torsion pendulum with identical masses. [A,An,E][7]
4. Measure the Young's modulus of a metallic scale using Cantilever oscillations. [A,An,E][7]
5. Study the effect of impurities on surface tension using capillary rise method. [A,An][7]
6. Study the variation of viscosity with temperature. [A,An][7]
7. Verify Bernoulli's theorem. [A,An][7]
8. Determine the coefficient of viscosity of a given viscous liquid by Stoke's method. [A,An,E][7]
9. Determine the velocity of ultrasonic waves in a liquid. [A,An,E][7]
10. Using a sonometer determine the frequency of a given tuning fork and verify the laws of strings. [A,An,E][7]
11. Write a python program to create an animation of the wave ($Asinkx - \omega t$). Make necessary changes in the program to animate a wave having $(Asinkx - \omega t) + (Asinkx + \omega t)$. [A,An][7]
12. Write a python program to model the deformation of the material under an applied load. Also generate a plot of the stress-strain relationship. [A,An][7]